1. **entify Basic Steps**:
   - Load A into R1.
   - Load B into R2.
   - Square the value in R1.
   - Multiply the squared value in R1 with A.
   - Multiply the value in R2 with B.
   - Add the results of the previous two operations.
   - Add C to the result.
   - Output the final result.
2. **Sequence of Steps**:
   - Step 0: Load A into R1.
   - Step 1: Load B into R2.
   - Step 2: Square the value in R1.
   - Step 3: Multiply the squared value in R1 with A.
   - Step 4: Multiply the value in R2 with B.
   - Step 5: Add the results of steps 3 and 4.
   - Step 6: Add C to the result.
   - Step 7: Output the final result and set Done=1.
3. **Create Table**:
4. **Truth Table**:
5. **Number of Flip-flops**:

From the truth table, we observe that we need at least 3 flip-flops to represent the states. These flip-flops will be used to store the current step, ensuring the FSM progresses through the correct sequence of steps.

This design ensures the computation is performed efficiently with the minimum number of steps and flip-flops required.

Step	Load R1	Load R2	Square R1	Multiply R1 with A	Multiply R2 with B	Add Results	Add C	Output	Done	Next Step
0	1	0	0	0	0	0	0	0	0	1
1	0	1	0	0	0	0	0	0	0	2
2	0	0	1	0	0	0	0	0	0	3
3	0	0	0	1	0	0	0	0	0	4
4	0	0	0	0	1	0	0	0	0	5
5	0	0	0	0	0	1	0	0	0	6
6	0	0	0	0	0	0	1	0	0	7
7	0	0	0	0	0	0	0	1	1	-

Step	Operation	R1	R2	Control Signals	Next Step
0	Load A into R1	A	-	Load R1, Mux1: A	1
1	Load B into R2	A	B	Load R2, Mux2: B	2
2	Square the value in R1	A2*A*2	B	ALU: Square, Mux1: R1	3
3	Multiply R1 with A	A2*A*2	AB	ALU: Multiply, Mux1: R1, Mux2: A	4
4	Multiply R2 with B	A2*A*2	B2*B*2	ALU: Multiply, Mux1: R2, Mux2: B	5
5	Add results of steps 3 & 4	A2*A*2	B2*B*2	ALU: Add, Mux1: R1, Mux2: R2	6
6	Add C to the result	A2*A*2	B2*B*2	ALU: Add, Mux1: R1, Mux2: C	7
7	Output the result	A2*A*2	B2*B*2	Output, Done=1	-